

		Simulation (L0 · 30 seeds)		Real Robots (L1 · N runs)		
Agents (n)	Occ.	Success Rate (mean ± std)	Makespan (mean ± std)	Succ. Rate† (mean ± std)	Makespan (mean ± std)	Runs (N)
4	0.16	0.992 ± 0.045	5.5 ± 1.5	0.950 ± 0.100	5.6 ± 0.8	5
5	0.20	0.987 ± 0.050	6.0 ± 1.3	0.950 ± 0.150	5.1 ± 0.8	10
6	0.24	0.978 ± 0.057	5.8 ± 1.6	1.000 ± 0.000	5.0 ± 1.1	5
7	0.28	0.933 ± 0.109	6.1 ± 1.3	0.900 ± 0.200	5.8 ± 2.7	5
8	0.32	0.938 ± 0.106	7.1 ± 1.8	0.870 ± 0.208	7.0 ± 5.2	27
9	0.36	0.933 ± 0.130	6.6 ± 1.7	0.778 ± 0.274	8.8 ± 5.3	5
10	0.40	0.933 ± 0.094	6.9 ± 1.5	0.693 ± 0.305	9.3 ± 6.6	7

Figure 2. PIBT scalability on a 5x5 grid (400 mm/cell, 25 cells): simulation L0 (30 random seeds per configuration, up to 100 steps) vs. real DotBot robots L1 (N physical runs). Success Rate : fraction of robots reaching their goal. Makespan : number of PIBT steps until the last robot arrives. Runs (N): total number of physical runs conducted for that agent count.